

Checkpoint A | Version 2 | Solutions

Each question earns *Pass* or *Not Yet*. *Pass (high)* and *Pass (low)* mark the range of passing work; both earn the same *Pass*.

Q1 | Opportunity Cost (A1.1)

Suppose Arthur can make 20 cakes *or* 10 drinks in one day.

What to look for:

Pass (high): Both per-unit tradeoffs and the best alternative forgone for each choice.

Pass (low): Correct ratios with an arithmetic slip, or an omitted unit made clear by the work.

Not Yet: Reversed tradeoffs, adding all alternatives, or missing evidence of the best alternative.

Grading note: On this checkpoint, a response with either parts (a)–(b) or part (c) correct earned Pass. The wording was less clear than intended, so we resolved that in the student's favor. Future assessments of this skill expect both.

a) What is Arthur's opportunity cost of 1 cake? **1/2 drink**

■ a) **1/2 drink:** giving up 10 drinks buys 20 cakes, so one cake costs 10/20 drinks.

b) What is Arthur's opportunity cost of 1 drink? **2 cakes**

■ b) **2 cakes,** the reciprocal.

c) Arthur's options for the day, best first: work in the shop, de-gnome the garden, take a Ministry shift.

(i) What is his opportunity cost of working in the shop? **De-gnoming the garden**

■ c) (i) De-gnoming the garden, his next best option after the shop.

(ii) What is his opportunity cost of the Ministry shift? **Working in the shop**

■ c) (ii) Working in the shop, his best alternative to the shift. The Ministry shift, ranked last, is the cost of nothing.

Checkpoint A | Version 2 | Solutions

Q2 | Arthur's PPF (A1.2)

Suppose Arthur can make 20 cakes *or* 10 drinks in one day. There are two graphs, one beside (a) and one beside (b). On both graphs, draw Arthur's PPF, drinks on the vertical axis and cakes on the horizontal. Label the intercepts.

What to look for:

Pass (high): A meaningful PPF, inside/on/outside classifications, and the effect of each capacity change.

Pass (low): An isolated wrong or missing label when the graph and other answers establish the distinction.

Not Yet: No usable PPF graph, repeated category confusion, or a wrong capacity-change answer without supporting evidence.

Grading note: On this checkpoint, a response with either part (a) or part (b) correct earned Pass. The wording was less clear than intended, so we resolved that in the student's favor. Future assessments of this skill expect both.

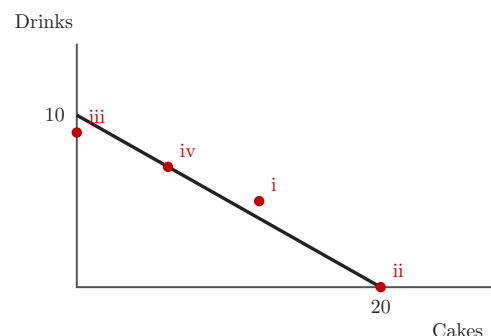
a) For each daily output, write whether it is unattainable, inefficient, or efficient, and mark it on the first graph.

(i) 12 cakes and 5 drinks **Unattainable**

(ii) 20 cakes **Efficient**

(iii) 9 drinks **Inefficient**

(iv) 6 cakes and 7 drinks **Efficient**



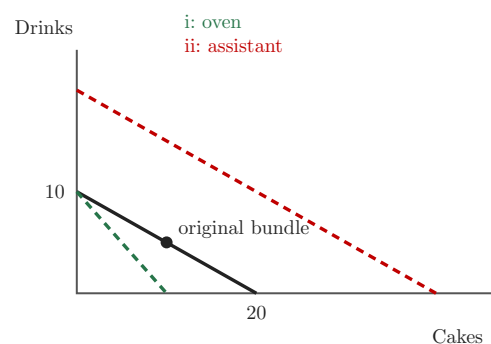
PPF: Straight line from 10 drinks to 20 cakes. Opportunity cost: 1/2 drinks per cake. a) **In order:** Unattainable; Efficient; Inefficient; Efficient.

b) Arthur currently makes 10 cakes and 5 drinks. On the second graph, draw his PPF and mark this output. Is that output unattainable, inefficient, or efficient:

(i) if his oven breaks down, halving his daily output of cakes?

Unattainable

(ii) if instead he hires an assistant, doubling the labor that goes into these two foods? **Inefficient**



b) The original bundle is on the original frontier. (i) Oven: cake intercept falls to 10, drinks stay at 10; **unattainable**. (ii) Assistant: both intercepts double; the original bundle is **inefficient**. **Taylor's Note:** In Q2.b, new frontiers illustrate the shifts but they are not required in the answer.

Checkpoint A | Version 2 | Solutions

Q3 | Absolute and Comparative Advantage (A2.1)

Luna and Neville both operate candy shops specializing in Tarts and Chocolates. In one day, Neville can make 10 tarts or 8 chocolates, and Luna can make 8 tarts or 2 chocolates.

What to look for:

Pass (high): Production and cost tables; AA from greater output and CA from lower opportunity cost.

Pass (low): A copying or arithmetic slip and its consistent consequences when the method and comparisons are clear.

Not Yet: Using output to choose CA, systematically reversing costs, or wrong answers with no cost work to show where the error came from. Correct answers earned Pass even without the cost work shown.

Set up a Production Table with both shops' output per day.

	Tart	Chocolate
Neville	10 Tarts	8 Chocolates
Luna	8 Tarts	2 Chocolates

Production table: each shop's output per day if it makes only that good, straight from the setup.

Then set up an Opportunity Cost table with both shops' opportunity costs for each good.

	Tart	Chocolate
Neville	$\frac{4}{5}$ Chocolate	$\frac{5}{4}$ Tarts
Luna	$\frac{1}{4}$ Chocolate	4 Tarts

Opportunity costs: each cell is what the shop gives up of the other good. Neville: a tart costs $\frac{8}{10} = \frac{4}{5}$ chocolate, a chocolate costs $\frac{10}{8} = \frac{5}{4}$ tarts. Luna: a tart costs $\frac{2}{8} = \frac{1}{4}$ chocolate, a chocolate costs $\frac{8}{2} = 4$ tarts.

Which shop has the absolute advantage (AA) and comparative advantage (CA) in each good?

AA in Tarts: **Neville**

CA in Tarts: **Luna**

AA in Chocolates: **Neville**

CA in Chocolates: **Neville**

AA in tarts: **Neville** ($10 > 8$). AA in chocolates: **Neville** ($8 > 2$).

CA in tarts: **Luna** ($\frac{1}{4} < \frac{4}{5}$). CA in chocolates: **Neville** ($\frac{5}{4} < 4$). Neville is better at both, but each shop holds exactly one comparative advantage.

Checkpoint A | Version 2 | Solutions

Q4 | Trade Between the Twins (A3.1)

Fred and George each make Fireworks and Pastilles for their joke shop. Their opportunity costs are listed in the following opportunity cost table.

	Pastille	Firework
George	1/5 Firework	5 Pastilles
Fred	1/2 Firework	2 Pastilles

What to look for:

Endpoints: The bounds identify an open interval. At an endpoint one producer is indifferent.

Pass (high): The lower-cost specialist, the full trade interval in common units, and an acceptable rate.

Pass (low): Correct specialist, bounds, and acceptable selection plus one extra outside checkbox.

Not Yet: Wrong specialization or bounds without correct work, only outside rates selected, or multiple extra selections.

a) If the twins specialize and trade with each other, which one should specialize in Pastilles? **George**

a) **George**, the twin with the lower opportunity cost of a pastille. Fred has the lower cost of a firework (2 versus 5 pastilles) and George the lower cost of a pastille (1/5 versus 1/2 firework).

b) What is the range of exchange rates that would make both Fred and George better off?

Between **2** and **5** Pastilles for 1 Firework

b) Between **2** and **5** pastilles for 1 firework: the two twins' opportunity costs of a firework, read down the Firework column. Below the low end the firework maker would rather keep his fireworks; above the high end the pastille maker would rather make fireworks himself.

c) Select all the exchange rates which could facilitate a trade between the twins.

☐ 1 Firework for 7 Pastilles ☐ 1 Firework for 6 Pastilles ☒ 1 Firework for 3 Pastilles ☐ 1 Firework for 1 Pastille

c) Only **1 firework for 3 pastilles**. 1 is below Fred's cost of 2, so he keeps his fireworks; 6 and 7 are above George's cost of 5, so he would rather make fireworks himself.